

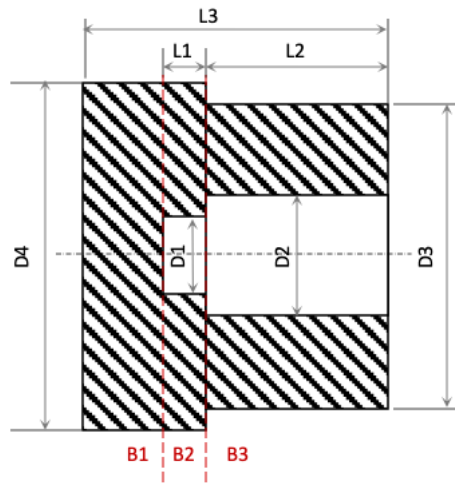
SOLUTIONS TO NUMERICAL PROBLEMS

METAL CASTING

The figure shows an axisymmetric casting to be fabricated by an expendable mould process.

Data: $D_1 = 50$ mm, $D_2 = 80$ mm, $D_3 = 200$ mm, $D_4 = 230$ mm, $L_1 = 30$ mm, $L_2 = 120$ mm, $L_3 = 200$ mm.

- 1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).
- 2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?
- 3) Considering: the axis of the casting as parting line, a volume of the casting cavity equal to $6,5 \text{ dm}^3$, a riser with a volume equal to 3 dm^3 , a middle gating system with H_{av} equal to 100 mm and $c = 0,6$. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.



Solution

- 1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).

In order to find the core dimensions, we need to consider the material allowance needed for the following machining operations and the shrinkage allowance (1,4%). The result must be rounded to mm. Therefore, the core dimensions are:

$$\begin{aligned}
 D_1 &= (50 - 2 \cdot 2)(1 + 0,01) = 46 \cdot 1,01 = 46,46 \rightarrow D_1 = 46 \text{ mm} \\
 D_2 &= (80 - 2 \cdot 2)(1 + 0,01) = 76 \cdot 1,01 = 76,76 \rightarrow D_2 = 76 \text{ mm} \\
 L_1 &= (30 - 2 + 2)(1 + 0,01) = 30 \cdot 1,01 = 30,3 \rightarrow L_1 = 30 \text{ mm} \\
 L_2 &= (120 - 2 + 2)(1 + 0,01) = 120 \cdot 1,01 = 121,2 \rightarrow L_2 = 121 \text{ mm}
 \end{aligned}$$

- 2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?

Element B1:

$$\begin{aligned}
 V_{B1} &= \frac{\pi}{4} D_4^2 \cdot (L_3 - L_1 - L_2) = \frac{\pi}{4} 230^2 \cdot (200 - 30 - 120) = \frac{\pi}{4} 230^2 \cdot 50 = 2.077.378 \text{ mm}^3 \\
 S_{B1} &= \frac{\pi}{4} D_4^2 + \frac{\pi}{4} D_1^2 + \pi D_4 \cdot (L_3 - L_1 - L_2) = \frac{\pi}{4} 230^2 + \frac{\pi}{4} 50^2 + \pi \cdot 230 \cdot 50 = 79.639 \text{ mm}^2 \\
 M_{B1} &= \frac{V_{B1}}{S_{B1}} = \frac{2.077.378}{79.639} = 26,1 \text{ mm}
 \end{aligned}$$

Element B2:

$$V_{B2} = \frac{\pi}{4}(D_4^2 - D_1^2) \cdot L_1 = \frac{\pi}{4}(230^2 - 50^2) \cdot 30 = 1.187.522 \text{ mm}^3$$

$$S_{B2} = \pi D_4 \cdot L_1 + \pi D_1 \cdot L_1 + \frac{\pi}{4}(D_4^2 - D_3^2) + \frac{\pi}{4}(D_2^2 - D_1^2)$$

$$S_{B2} = \pi \cdot 30 \cdot (230 + 50) + \frac{\pi}{4}(230^2 - 200^2) + \frac{\pi}{4}(80^2 - 50^2) = 39.584 \text{ mm}^2$$

$$M_{B2} = \frac{V_{B2}}{S_{B2}} = \frac{1.187.522}{39.584} = 30 \text{ mm}$$

Element B3:

$$V_{B3} = \frac{\pi}{4}(D_3^2 - D_2^2) \cdot L_2 = \frac{\pi}{4}(200^2 - 80^2) \cdot 120 = 3.166.725 \text{ mm}^3$$

$$S_{B3} = \pi D_3 \cdot L_2 + \pi D_2 \cdot L_2 + \frac{\pi}{4}(D_3^2 - D_2^2) = \pi \cdot 120 \cdot (200 + 80) + \frac{\pi}{4}(200^2 - 80^2) = 131.947 \text{ mm}^2$$

$$M_{B3} = \frac{V_{B3}}{S_{B3}} = \frac{3.166.725}{131.947} = 24 \text{ mm}$$

Directional solidification:

The first element that will solidify is Element B3, then Element B1 and at last Element B2. Being Element B2 in between Elements B1 and B3, the riser has to be positioned on the Element B2 (last to solidify).

- 3) *Considering: the axis of the casting as parting line, a volume of the casting cavity equal to 6,5 dm³, a riser with a volume equal to 3 dm³, a middle gating system with H_{av} equal to 100 mm and c = 0,6. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.*

The pouring velocity in the bottleneck cross section (A_{bn}) will be

$$v = c \cdot \sqrt{2 \cdot g \cdot H_m} = 0,6 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{100}{1000}} = 0,840 \frac{\text{m}}{\text{s}} = 840 \frac{\text{mm}}{\text{s}}$$

Being the mould filling time equal to $T_{MF} = \frac{V_{total}}{A_{bn} \cdot v}$, we get:

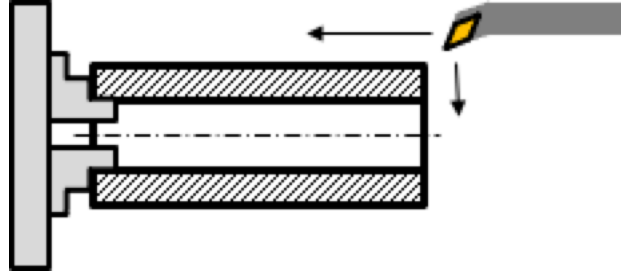
$$V_{total} = V_{casting} + V_{risers} = 6,5 + 3 = 9,5 \text{ dm}^3 = 9.500.000 \text{ mm}^3$$

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v} = \frac{9.500.000}{10 \cdot 840} = 1.131 \text{ mm}^2$$

TURNING

The finishing of a hollow cylindrical (tubular) steel component is carried out with the same tool by performing two operations (see figure):

- External cylindrical turning over the entire length of the workpiece.
- Facing the free base (right).



Using the following data, determine:

- The cutting speed in order to maximize the production rate.
- The maximum allowed feed in order to meet a surface finish requirement on the outer cylindrical surface equal to $R_a = 0,4 \mu\text{m}$.
- The minimum clamping pressure to properly hold the workpiece (the feed rate is $0,08 \text{ mm/rev}$, a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 120 mm^2 and friction coefficient equal to $0,35$).
- The contact time of the facing operation in the hypothesis of carrying out the machining with a constant number of revolutions per minute equal to 540 and a feed rate of $0,08 \text{ mm/rev}$. What is the value of the maximum cutting speed?

Data: Initial external diameter = 236 mm , final external diameter = 233 mm , initial length = 300 mm , internal diameter = 150 mm , $k_{c0,4} = 2100 \text{ MPa}$, $x = 0,29$, tool changing time = 120 s , tool radius = $0,8 \text{ mm}$, $k_r = 100^\circ$, $k'_r = 10^\circ$, Taylor's law coefficient: $C = 510$, $n = 0,35$.

Solution

- Determine the cutting speed in order to maximize the production rate.*

In order to maximize the production rate, we need to minimize the production time. Therefore:

$$T_{opt} = T_t \left(\frac{1}{n} - 1 \right) = \frac{120}{60} \left(\frac{1}{0,35} - 1 \right) = 3,714 \text{ min}$$

and:

$$v_{c,opt} = \frac{C}{T_{opt}^n} = \frac{510}{3,714^{0,35}} = 322,2 \text{ m/min}$$

- Determine the maximum allowed feed in order to meet a surface finish requirement on the outer cylindrical surface equal to $R_{a,limit} = 0,4 \mu\text{m}$*

Considering the Schmalz's equation (tool nose radius greater than zero), we have:

$$f = \sqrt{\frac{32 \cdot r \cdot R_{a,limit}}{1000}} = \sqrt{\frac{32 \cdot 0,8 \cdot 0,4}{1000}} = 0,1 \text{ mm/giro}$$

We need to check if the Schmalz's equation hypotheses hold:

$$\begin{cases} \frac{f}{2} \leq r \sin k_r & \left\{ \frac{0,1}{2} = 0,05 \leq 0,8 \cdot \sin 100^\circ = 0,7878 \rightarrow ok \right. \\ \frac{f}{2} \leq r \sin k'_r & \left\{ \frac{0,1}{2} = 0,05 \leq 0,8 \cdot \sin 10^\circ = 0,1389 \rightarrow ok \right. \end{cases}$$

- 3) Determine the minimum clamping pressure to properly hold the workpiece (the feed rate is 0,08 mm/rev, a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 120 mm² and friction coefficient equal to 0,35)

In order to check if the workpiece is properly held, the resistant torque should be greater than the cutting torque.

$$T_c = F_c \cdot \frac{D_f/2}{1000} \leq T_r = z \cdot \mu \cdot p \cdot A \cdot \frac{D_{clamping}/2}{1000}$$

Therefore, we need to compute the cutting force, as follows:

$$k_c = k_{c0,4} \cdot \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2100 \cdot \left(\frac{0,4}{0,08 \cdot \sin 100^\circ} \right)^{0,29} = 3364 \text{ MPa}$$

$$F_c = k_c \cdot f \cdot a_p = 3364 \cdot 0,08 \cdot 1,5 = 403,68 \text{ N}$$

$$T_c = F_c \cdot \frac{D_f/2}{1000} = 403,68 \cdot \frac{233/2}{1000} = 47 \text{ Nm}$$

As a consequence, we have the following minimum clamping pressure is needed:

$$p \geq p_{min} = \frac{T_c}{z \cdot \mu \cdot A \cdot \frac{D_{clamping}/2}{1000}} = \frac{47}{3 \cdot 0,35 \cdot 120 \cdot \frac{150/2}{1000}} = 4,97 \text{ MPa} \cong 5 \text{ MPa}$$

- 4) Determine the contact time of the facing operation in the hypothesis of carrying out the machining with a constant number of revolutions per minute equal to 540 and a feed rate of 0,08 mm/rev. What is the value of the maximum cutting speed?

The contact time can be determined as follows:

$$L_c = (D_f - D_{internal})/2 = (233 - 150)/2 = 41,5 \text{ mm}$$

$$t_c = \frac{L_c}{v_f} = \frac{L_c}{f \cdot n} = \frac{41,5}{0,08 \cdot 540} = 0,96 \text{ min} = 57,64 \text{ s}$$

Given the constant n , since during facing the speed varies with the diameter, the maximum cutting speed will occur in correspondence with the maximum diameter:

$$v_{c,max} = \frac{\pi \cdot D_f \cdot n}{1000} = \frac{\pi \cdot 233 \cdot 540}{1000} = 395,3 \text{ m/min}$$

MILLING

A steel planar surface with a length equal to 500 mm and a width equal to 50 mm must be flattened. A face milling roughing operation will remove a machining allowance equal to 5 mm. The face mill will be positioned so that 10% of the tool diameter will exit from one side of the plane width.

Determine:

- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 12 kW and an efficiency of 0,8.
- 2) The cutting time needed to carry out the operation.

Data: mill diameter 80 mm, number of cutting edges $Z = 7$, entering angle = 90° , $f_z = 0,15$ mm/rev, $k_{cs} = 1600$ MPa, $x = 0,197$.

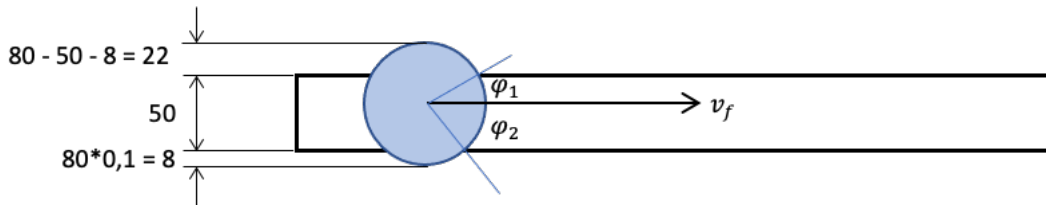
Cutting speed: if $f_z = 0,05$ mm/rev then $v_c = 305$ m/min; if $f_z = 0,1$ mm/rev then $v_c = 275$ m/min; if $f_z = 0,2$ mm/rev then $v_c = 225$ m/min.

Solution

- 1) *The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 12 kW and an efficiency of 0,8.*

In order to answer, it is necessary to understand what the actual situation in this face milling operation is.

First of all, we need to compute the engagement angle and then the number of simultaneously working teeth.



Referring to the figure, the overall engagement angle is the sum of the following two angles:

$$\varphi_1 = \arcsin\left(\frac{80/2 - 22}{80/2}\right) = 26,74^\circ$$

$$\varphi_2 = \arcsin\left(-\frac{80/2 - 8}{80/2}\right) = -53,13^\circ$$

$$\varphi = \varphi_1 + |\varphi_2| = 26,74^\circ + 53,13^\circ = 79,87^\circ$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{360}{7} = 51,43^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{79,87^\circ}{51,43^\circ} = 1,55$$

Therefore, we have to consider the worst-case scenario when two cutting edges are simultaneously cutting, that we have when the angular position of the two cutting edge is symmetric with respect to the feed direction, so:

$$\text{Cutting tooth 1, angular position: } \theta_1 = \frac{\varphi_0}{2} = \frac{51,43^\circ}{2} = 25,715^\circ \leq 26,74^\circ$$

$$\text{Cutting tooth 2, angular position: } \theta_2 = -\frac{\varphi_0}{2} = -\frac{51,43^\circ}{2} = -25,715^\circ \geq -53,13^\circ$$

The two angular position are coherent with respect to the engagement angle.

In order to do answer the question we notice that:

$$h_D(\theta_1) = f_z \cos \theta_1 = 0,15 \cdot \cos 25,715^\circ = 0,135 \text{ mm}$$

$$h_D(\theta_2) = f_z \cos \theta_2 = 0,15 \cdot \cos -25,715^\circ = 0,135 \text{ mm}$$

Therefore:

$$h_D(\theta_1) = h_D(\theta_2) = h_D$$

It follows:

$$A_D = h_D \cdot a_p = 0,135 \cdot 5 = 0,675 \text{ mm}^2$$

$$k_c = k_{cs} h_D^{-x} = 1.600 \cdot 0,135^{-0,197} = 2.374 \text{ MPa}$$

$$F_c = k_c A_D = 2.374 \cdot 0,675 = 1.602 \text{ N}$$

$$T_c = F_{c,\theta_1} \frac{D/2}{1000} + F_{c,\theta_2} \frac{D/2}{1000} = 2 \cdot F_c \frac{D/2}{1000} = 2 \cdot 1.602 \frac{80/2}{1000} = 128,2 \text{ Nm}$$

Being $f_z = 0,15$, from the data by linear interpolation we can choose the adequate cutting speed as:

$$v_c = \frac{275 + 225}{2} = 250 \frac{\text{m}}{\text{min}}$$

So:

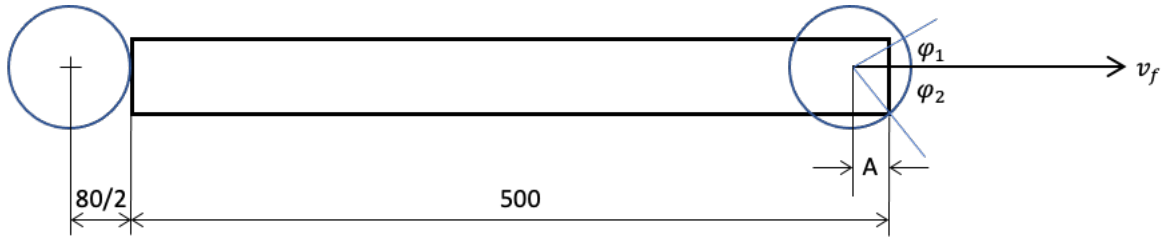
$$\omega = \frac{1000}{60} \cdot \frac{v_c}{D/2} = \frac{1000}{60} \cdot \frac{250}{80/2} = 104,2 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 128,2 \cdot 104,2 = 13.358 \text{ W} = 13,4 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,8 \cdot 12 = 9,6 \text{ kW}$$

Therefore, being $P_c = 13,4 \text{ kW} \geq P_{c,limit} = 9,6 \text{ kW}$, the face milling operation is not feasible.

2) The cutting time needed to carry out the operation.



Referring to the figure, the cutting time is equal to:

$$t_c = \frac{L_{workpiece} + \frac{D}{2} - A}{v_f} = \frac{500 + \frac{80}{2} - A}{v_f}$$

Where

$$A = \min \left\{ \frac{D}{2} \cos \varphi_1; \frac{D}{2} \cos \varphi_2 \right\} = \min \left\{ \frac{80}{2} \cos 26,74^\circ; \frac{80}{2} \cos 53,13^\circ \right\} = \min \{35,72; 24\} = 24 \text{ mm}$$

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 250}{\pi \cdot 80} = 995 \text{ RPM}$$

$$v_f = Z \cdot f_z \cdot n = 7 \cdot 0,15 \cdot 995 = 1.045 \frac{\text{mm}}{\text{min}}$$

Therefore:

$$t_c = \frac{500 + \frac{80}{2} - 24}{1.045} = 0,494 \text{ min} = 29,6 \text{ s}$$